

An experimental platform for investigating astrophysically relevant magnetized plasma jets on the COBRA facility

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ABSTRACT

A new platform has been developed for the 1-MA COBRA generator to investigate the physical processes affecting the formation, collimation, and stability of high-speed outflows in magnetically driven laboratory plasma jets. Such experiments serve as diagnostically accessible surrogates for astrophysical jets under the assumption that the underlying dynamics are scale invariant. In contrast to previous current driven high energy density laboratory jet experiments that use radial/conical wire arrays or foils, the platform described here uses azimuthally symmetric gas-puff injection. This avoids the ablation phase from a solid target, allowing the jets to develop earlier and be driven longer without depleting their mass source and disrupting. A permanent magnet provides an initial poloidal magnetic field, which links the two concentric electrodes and mimics the boundary conditions of a star-accretion disk system. Extended magnetohydrodynamic effects can be assessed using a polarity convolute, which allows for reversal of the electrode bias. **The resulting plasma jets exhibit remarkable stability, persisting for hundreds of nanoseconds and achieving aspect ratios $\geq 30:1$.**

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I. INTRODUCTION

Astrophysical jets¹ are collimated, high-speed outflows observed to be natural features of objects that spin and accrete matter. Developing over a vast range of scale lengths and source energies, jets are common features of planetary nebula, young stellar objects, neutron stars, and active galactic nuclei (AGN).^{2–4} These jets are generally much larger than the originating source-accretion disk system and range in length from hundreds of astronomical units to megaparsecs, making them some of the largest coherent objects in the universe. Outflow velocities generally scale with source mass and range from several hundred km/s for stellar mass sources to relativistic velocities for supermassive black holes. Protostellar jets are typically over-dense with respect to the ambient medium,⁵ while relativistic jets consisting of electron-positron plasmas are believed to be under-dense.⁶ Despite these differences, common features suggest that universal scale invariant mechanisms may be responsible for jet launching, collimation, and stability. While no single model is universally accepted, magnetic fields are widely believed to play a critical role.^{7–18}

In one prominent hydromagnetic model,¹⁴ differential rotation of an accretion disk twists up poloidal ($\hat{r}$, $\hat{z}$) magnetic fields, generating a

configuration that progresses through a series of force-free states satisfying the Beltrami condition $\nabla \times \mathbf{B} = \lambda \mathbf{B}$, where λ is a field line constant. In the presence of an effective dissipation mechanism, the eigenvalue λ is expected to be a global constant and such plasmas are said to be relaxed.^{19–21} In cylindrical geometry, azimuthally symmetric solutions (with azimuthal mode number $m = 0$)²² become energetically unfavorable with continued twisting of the field lines.²³ Therefore, any steady state force-free jet is expected to be helical with both $m = 0$ and $m = 1$ components.^{21,24} In this state, continued winding at the source serves to increase the amplitude of the helical mode, not the net axial current. Notably, magnetized double-helical structures resembling such a configuration have recently been reported in high-fidelity radio images of AGN jets,²⁵ as well as laboratory planar plasma gun experiments.²⁶

These and other magnetized jet models generally use the framework of ideal magnetohydrodynamics (MHD) due to the dominating dynamo driven electric fields in Ohm's law. However, astrophysical observations²⁷ and laboratory experiments^{28–30} are increasingly revealing the importance of extended-MHD effects, such as the Hall terms, even when such terms may only be significant in the low-density

periphery where scales are of the order of the ion inertial length. To properly consider the simultaneous dynamics of flows and magnetic fields, more generalized models of plasma relaxation and self-organization have been developed.^{31–35} As opposed to MHD force-free states which satisfy a single Beltrami condition for the magnetic field, these generalized models predict multi-Beltrami states that permit solutions with dynamically significant thermal pressures and stabilizing shear flows.^{31,36} Furthermore, with the inclusion of only Hall terms, such models establish the foundation of a unified dynamo-reverse dynamo mechanism that may be directly applicable to astrophysical outflows.^{37,38}

Testing these models through astrophysical observation alone is extremely challenging, as such observations are necessarily remote and quasistatic due to the long dynamical timescales of these systems. Another approach is, therefore, needed to resolve the long-term dynamics and underlying physical processes of jet formation, propagation, and stability. This has motivated experimentalists over the last several decades to reproduce astrophysically relevant plasma jets in a laboratory environment. Early experiments based on laser ablation of foils and foams^{39–41} produced purely hydrodynamic jets and focused on the detailed effects of high Mach number flows, shocks, and radiative cooling. However, more recent laser driven experiments have since demonstrated the ability to produce collimated jets with self-generated mega-Gauss fields.⁴² Other experiments driven by mega-Ampère current generators have traditionally used radial or conical wire arrays or foils to produce hydrodynamic precursor jets followed by z-pinch like jets confined within a magnetic cavity (see⁴³ for a review). The magnetized jets develop fairly late into the current pulse after the solid has ablated and are, therefore, analogous to the implosion phase of a wire-array z-pinch. Such jets extend ~ 1 – 2 cm before instabilities break the central column. In some cases, episodic jets are observed with multiple successive jets forming within their distinct magnetic bubbles.⁴⁴ More recently, medium energy density experiments using azimuthally continuous gas-puff injection and three concentric planar electrodes⁴⁵ have produced stable, high-aspect ratio ($>25:1$) plasma jets with strong helical shear flows⁴⁶ embedding a double-helix magnetic structure.²⁶ The spontaneous formation of such complex magnetic and flow structures suggests a generalized relaxation process that may also be operating in naturally occurring astrophysical plasma jets.

To explore this hypothesis that such jets are stable, self-organized, driven equilibria governed by a more generalized model of plasma relaxation—and to study the scaling of these phenomena to higher energy densities—a new experimental platform has been developed for the 1-MA COBRA generator. This platform is designed to produce magnetically driven, self-organized jets by replicating the boundary conditions and dimensionless scaling relations of jets from young stellar objects. In this paper, we present the design of the new platform (Sec. II) as well as preliminary experimental observations (Sec. III). We conclude with summary of the results and a discussion of future work (Sec. IV).

II. DESIGN

The new experimental platform combines the approach of previous medium energy density planar plasma gun experiments^{45–49} with the pulsed-power and diagnostic capabilities of the 1-MA COBRA generator. These plasma guns are designed to capture the dynamics of astrophysical jets by replicating similar boundary conditions (Fig. 1) and dimensionless scaling relations, with sufficiently large or small

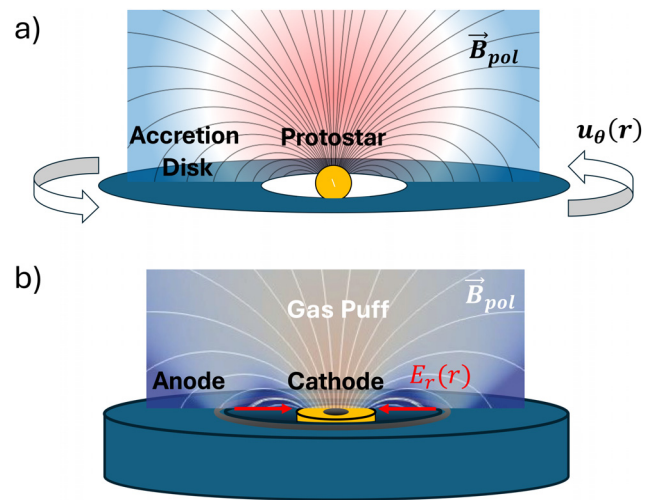


FIG. 1. (a) General boundary conditions of a magnetic tower jet consisting of magnetized central engine surrounded by a differentially rotating accretion disk. Poloidal field lines ($\vec{B}_{pol}$) frozen to the accretion disk are twisted, producing a radial electric field that drives a current between the disk and central engine. As the differential rotation continues, the magnetic pressure associated with the growing toroidal field leads to collimation and expansion of a jet the axial direction. (b) Electrode configuration designed to mimic the boundary conditions off the magnetic tower jet by providing a radial electric field in the presence of a poloidal magnetic field. An azimuthally continuous gas-injection slit allows for free rotation of the jet footpoint with $u_\theta = -E_r/B_z$.

dimensionless numbers (i.e., Reynolds, magnetic Reynolds, and Péclet). Here, sufficiently large or small means that both systems are well described by the ideal MHD equations and effects of viscosity, magnetic field diffusion, and thermal conduction are negligible on a global scale. Because the ideal MHD equations have no intrinsic scale length, laboratory and astrophysical plasma jets in this regime should evolve similarly if times scale as $a\sqrt{b/c}$, velocities as $\sqrt{c/b}$, and magnetic fields as $\sqrt{c}$, where a is the ratio of jet radii, b is the ratio of mass densities, and c is the ratio of plasma pressures.⁵⁰ Our experimental platform was, therefore, designed to satisfy these scaling relations for protostellar jets to within an order of magnitude (Table I).

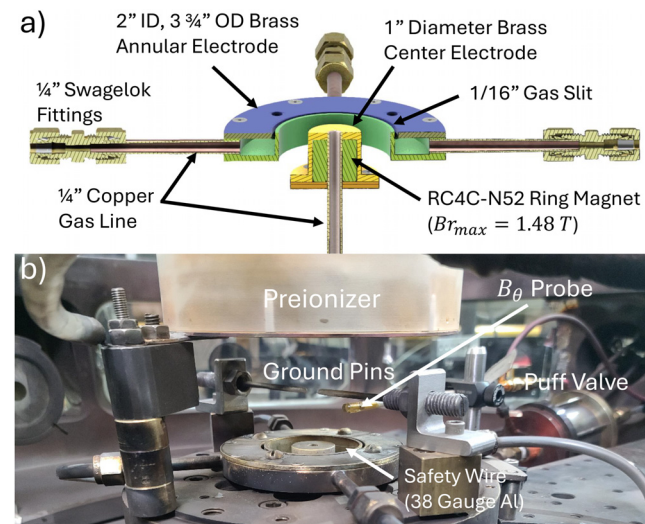
As opposed to several recent pulsed-power experiments on mega-Ampère drivers that attempt to replicate the boundary conditions of a spinning accretion disk using wire-array ablation flows imparted with an initial angular momentum,^{51,52} our platform uses two fixed, concentric, planar electrodes to establish a radial electric field that is roughly equivalent to an accretion disk rotating azimuthally in a poloidal magnetic field (Fig. 1). Here, the radial electric field in the presence of poloidal magnetic field is expected to drive rotation according to $u_\theta = -E_r/B_z$, and not vice versa. In this way, the electrodes mimic the boundary conditions of a magnetic tower¹⁴ or tower-like jet⁴⁹ and are similar to previous radial foil or wire array experiments.^{29,44,53–57} A major difference, however, is that we use direct gas-puff injection as opposed to a solid target. This injection system is analogous to those used previously in larger planar plasma gun experiments (e.g., Fig. 1 of Ref. 47); however, instead of discrete holes, we adopt azimuthally continuous sources, similar to those used in Ref.⁴⁵ which allow for free rotation of the jet foot-points. The use of

TABLE I. Comparison of experimental and astrophysical jet parameters using MHD scaling relations.⁵⁰

Parameter	YSO Jet ^a	MOCHILabjet ^b	COBRA Jet	YSO/COBRA	MOCHI/COBRA	ratio symbol
Radius [cm]	10^{15}	5	0.15	7×10^{15}	33	<i>a</i>
Mass Density [g cm^{-3}]	10^{-19}	10^{-7}	6×10^{-5}	1.7×10^{-15}	2×10^{-3}	<i>b</i>
Pressure [Pa]	10^{-8}	4×10^4	8×10^6	1.3×10^{-15}	5×10^{-3}	<i>c</i>
Dynamical Age [ns]	10^{19}	5×10^4	500	2×10^{16}	100	<i>d</i>
Expansion Velocity [km s^{-1}]	300	100	160	1.9	0.6	<i>e</i>
Magnetic Field [T]	5×10^{-7}	0.5	10	10^{-8}	0.05	<i>f</i>
Length [cm]	10^{18}	110	7.2	1.4×10^{17}	15	
Temperature [eV]	1	5	50	0.02	0.1	
Sound Speed [km s^{-1}]	10	30	35	0.3	0.9	
Alfvén Speed [km s^{-1}]	40	50	35	1.14	1.4	
Reynolds	$>10^8$	$\sim 10^4$	$\sim 10^5$	$>10^3$	0.1	
Magnetic Reynolds	$>10^{15}$	$\sim 10^3$	~ 70	$>10^{13}$	14	
Peclet	$>10^7$	$\sim 10^3$	~ 50	$>10^5$	20	
β	0.01–100	~ 0.2	~ 1	0.01–100	0.2	
MHD scaling: Time				0.4	0.2	$a\sqrt{b/c/d}$
MHD Scaling: Velocity				0.5	2.8	$\sqrt{c/b/e}$
MHD Scaling: Magnetic Field				0.7	1.4	$\sqrt{c/f}$

^aOrder of magnitude estimates for YSO jets obtained from Refs. 5, 6, and 81–84.^bMOCHILabjet values from Refs. 45 and 46.

gas avoids the ablation stage and formation of a hydrodynamic precursor jet, enabling the magnetically driven jets to develop at the start of the current pulse instead of at the end. Another design difference is that we use a permanent magnet to provide the dipole field linking the two electrodes as opposed to dynamic fields produced by twisted return current posts^{29,54} or quasi-static fields from external field coils.^{45,47,56,57} This approach simplifies the design but lacks the controllability of electromagnetic coils.

**FIG. 2.** (a) Design of the electrode assembly. (b) Picture of the electrode assembly mounted in the COBRA vacuum chamber for an experimental shot campaign. Here, the electrodes are mounted without the use of a polarity convolute.

The electrode design and experimental setup is shown in Fig. 2. The electrodes are machined in house from brass and consist of an annular electrode and a central top-hat electrode. The annular electrode has a 2" inner diameter and 3.75" outer diameter and consists of two pieces: a main body, and a lid which establishes an internal reservoir and 1/16" annular slit at the inside radius. Four 1/4" copper gas lines with Swagelok fittings are brazed to through holes in the main body. The reservoir allows the gas to spread out before exiting the slit, improving azimuthal symmetry. The central electrode also consists of two pieces, a brass disk with a diameter of 1.5" brazed to a 1/4" copper gas line and a brass top-hat that reduces to a diameter of 1". The top-hat encloses a RC4C-N52 nickel-plated neodymium ring magnet with a residual induction of 1.48 T. Dipole field orientations that point from the annular electrode into the central electrode are referred to as $-B_z$ while field orientations that point from the central electrode to the annular electrode are $+B_z$. Experimental shots can also be taken without the magnet in place ($0 B_z$). During experiments, two 1/8" tungsten rods are mounted above the electrodes and are kept at the same potential as the annular electrode. To lower the initial impedance and reduce unacceptably high load voltages and power reflections, two safety measures are implemented. First, the gas is pre-ionized by an independent -6.5 kV source prior to firing COBRA. Second, an additional "safety wire" (typically 38 gauge aluminum) connects the two electrodes.

Gas is provided to each electrode using two identical custom fast gas-puff valves (Fig. 3). The valve pancake solenoids (18 gauge, 20 turn magnet wire) are driven by the same power supply, and therefore, all the gas lines are the same length (19.5"). The valve feeding the annular electrode is branched into four independent lines which feed each quadrant of the electrode. Care is taken to ensure that the valves are inductively isolated during a COBRA pulse. Valve performance was characterized using planar laser-induced fluorescence (PLIF)⁵⁸ [Fig. 3(b)], which

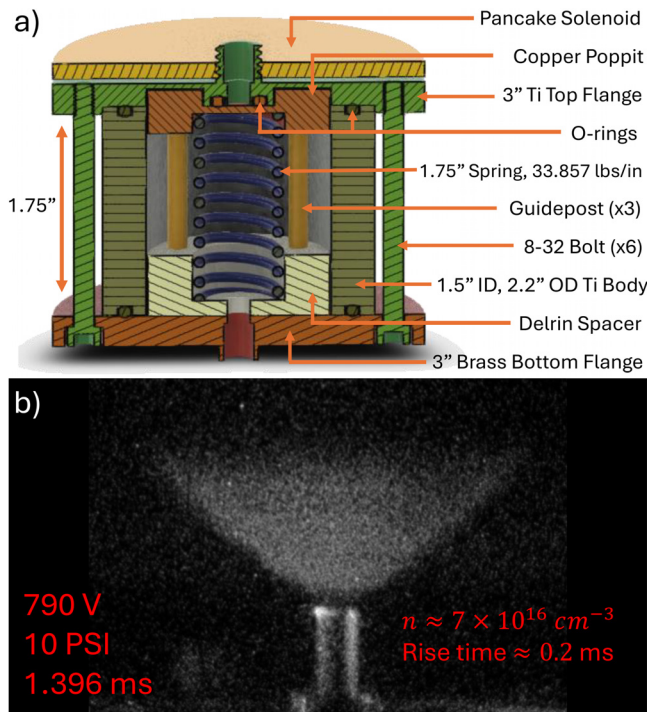


FIG. 3. (a) Design of custom fast gas-puff valve which provides gas to the two electrodes. (b) Planar laser-induced fluorescence image taken during characterization of the fast gas-puff valve 1.396 ms after triggering the valve solenoid. The gas plume from the 1/4" gas line expands at a 45° angle, as expected for choked flow.

demonstrated that for nominal backfill pressures and operational voltages the puff rise time was of the order of 0.2 ms with densities approaching 10^{17} cm^{-3} . Gas is injected just prior to a COBRA pulse such that there is a minimum delay between successful triggering of the preionizer and current start. This ensures that the gas-puff is localized above the electrodes and the chamber is not pre-filled (e.g., Fig. 5 in Ref. 59).

While our laboratory jet dimensionless numbers are sufficient to ensure the bulk plasma is in the ideal MHD regime, they still differ by many orders of magnitude from their astrophysical counterparts. This implies that the scale length at which the ideal MHD description becomes invalid is much larger in proportion to the global scale for our laboratory jets. Extended MHD effects are therefore expected to play a more significant role in their dynamics. Recent experiments support this conjecture, demonstrating that even in situations where the bulk plasma is well described by ideal MHD, extended MHD effects in the low-density periphery can impact the global dynamics.^{28–30} To investigate these effects, our experimental platform can be easily integrated with a polarity reversing convolute (Fig. 4). This enables us to probe the influence of the Hall term, not present in ideal MHD, which captures an inherent asymmetry based on the direction of current flow. This asymmetry is revealed in a simplified version of the generalized Ohm's law,²⁸

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} - \frac{1}{en_e} \mathbf{j} \times \mathbf{B} = \eta \mathbf{j}, \quad (1)$$

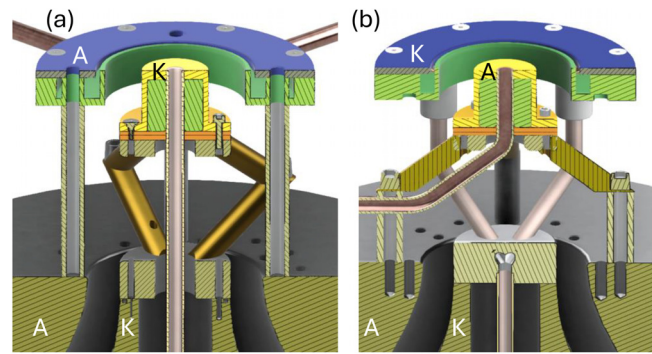


FIG. 4. Experimental hardware mounted onto a polarity reversing convolute in standard polarity (a) and reverse polarity (b). In reverse polarity, the annular electrode is modified to allow gas lines to enter from the bottom. Both configurations are designed to have a similar inductance.

where e is the electron charge, n_e is the electron density, j is the current density, and η is the plasma resistivity. The Hall term (term 3) generates an electric field perpendicular to the resistive electric field when currents flow perpendicular to magnetic fields. To isolate this effect from other potential confounding variables, the convolute is designed to have very little difference in inductance based on the direction of current. For standard polarity (SP), the current flows radially inward from the annular electrode and for reverse polarity (RP) it flows outward. The reverse polarity configuration requires a modified annular electrode body to enable gas delivery from four copper gas-line legs.

III. INITIAL OBSERVATIONS

Here, we present preliminary measurements from the first three shot campaigns on the COBRA generator. In the first campaign, the electrodes were placed in standard polarity (SP) without the use of any convolute hardware. For the first several shots, different safety wire and ground pin configurations were tested until a pronounced jet was observed using the configuration shown in Fig. 2(b). Here, the tungsten ground pins were located 26 mm above the electrodes and were gradually opened (on axis) throughout the campaign to a maximum gap of ~ 20 mm to allow the plasma jets to pass through. The preionizer was mounted 55 mm above the electrode plane. For the second and third shot campaigns, the electrodes were mounted on the polarity convolute in standard and reverse polarity (respectively), with tungsten pins 45 mm above the electrode plane and the preionizer 80 mm above. For all the shots presented here, argon was used as the working gas with a backfill pressure of 15 psia in each valve. The driver current delivered to the load hardware is shown in Fig. 5 for various shots in each campaign and is reasonably consistent, particularly during the initial rise (< 250 ns). Current crowbarring, where the current is sustained after its peak, was observed in many shots during the first and third campaigns. This behavior is often indicative of arcing within the magnetically insulated transmission line (MITL) or insulating ring stack. It is not immediately clear why this crowbarring appears to be absent in the second shot campaign.

Figure 6 shows the evolution of representative plasma jets from each campaign using visible and XUV self-emission imaging. During the early stages of jet formation (≤ 100 ns), the plasma resembles the final implosion stage of a dense plasma focus as the current linking the

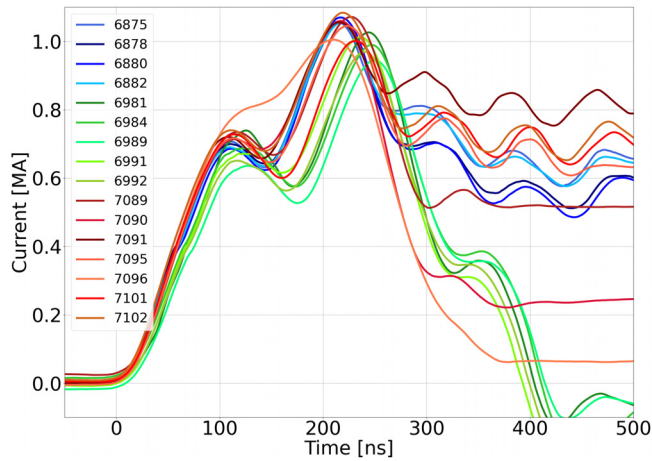


FIG. 5. COBRA current measured using a Rogowski coil around the generator cathode for various shots in each of the three shot campaigns (campaign 1 - blues, campaign 2 - greens, campaign 3 - reds).

center and annular electrode gas-puffs causes the flared central column to pinch, collimate, and drive axial flows (see⁴⁹ for a comprehensive theoretical description). Ablation of the safety wire is observed to varying degrees, as well as kinking of the central column, presumably due

to a current driven kink instability.⁶⁰ It is during this initial pinching phase that x-rays are detected using a photoconducting diamond diode (PCD) with a 6 μm Mylar filter ($\geq 10\%$ transmission for photon energies ≥ 1 keV). As the shot progresses, the central column lengthens (zippers) while maintaining an approximately constant radius. In most cases, small helical perturbations or knots are observed to propagate axially with the jet without growing substantially in the radial direction over the observation window. These features may be indicative of saturated kink modes⁶¹ or internal shock structures.⁶²

When plotting the jet length as a function of time using visible self-emission images (Fig. 7), a distinct difference is observed between standard and reverse polarity electrode orientations. For standard polarity (SP) shots, the base of the jet pinches ~ 60 ns into the current pulse and zippers up the axis at ~ 150 km/s. For reverse polarity (RP) shots, the jets pinch ~ 50 ns later and lengthen at ~ 170 km/s. In both cases, the expansion velocity is remarkably constant, indicating very little acceleration or deceleration of the leading edge of the jet as it lengthens. This is consistent with previous ideal MHD simulations of laboratory magnetic tower jets.⁶³ Due to the consistency of the driver current across different polarities, a disparity in gas puff valve performance between the two otherwise identical valves is likely the primary factor influencing the different jet trajectories. This reasoning follows from theoretical considerations discussed below as well as the fact that the same valves were used in their respective locations throughout the shot campaigns, such that a polarity reversal swaps the valve connected

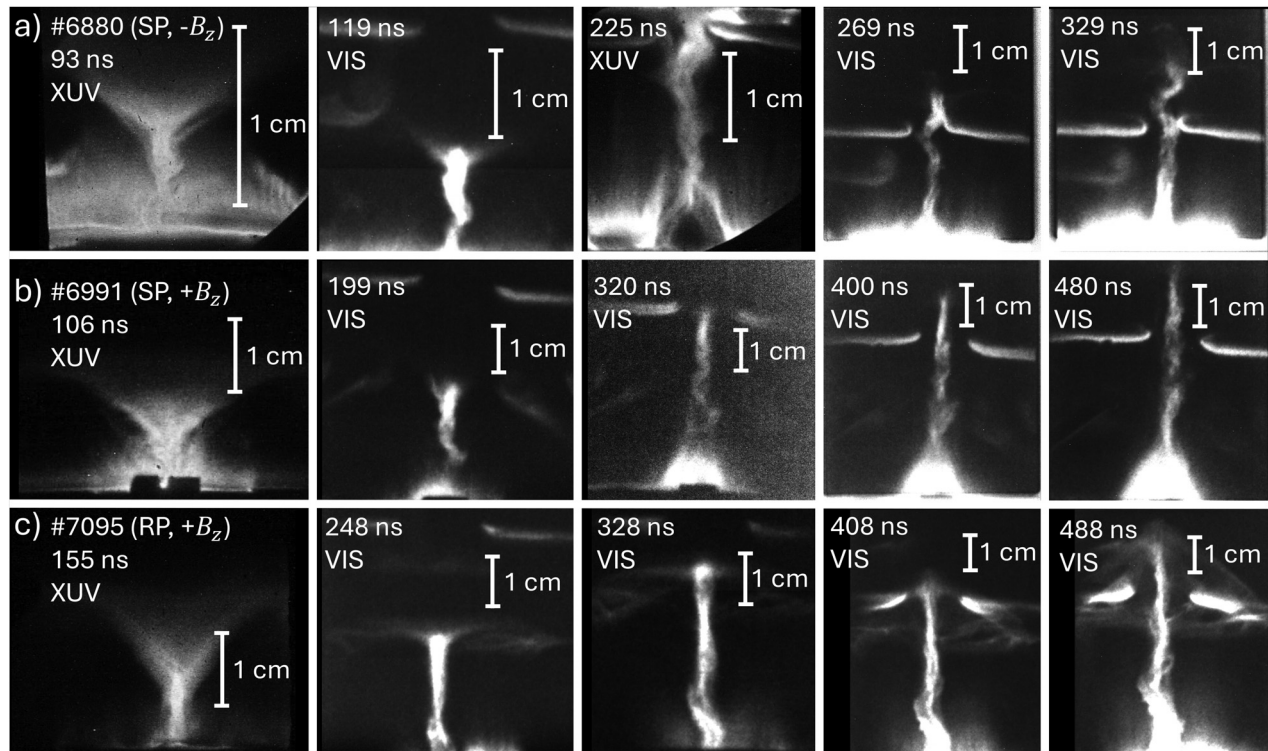


FIG. 6. Time evolution of various jets in each of three shot campaigns on COBRA. (a) Shot #6880 in standard polarity without the polarity convolute. The poloidal magnetic field points into the central electrode ($-B_z$). (b) Shot #6991 in standard polarity with the polarity convolute. The poloidal magnetic field points out of the central electrode ($+B_z$). (c) Shot #7095 in reverse polarity. The poloidal magnetic field points out of the central electrode ($+B_z$).

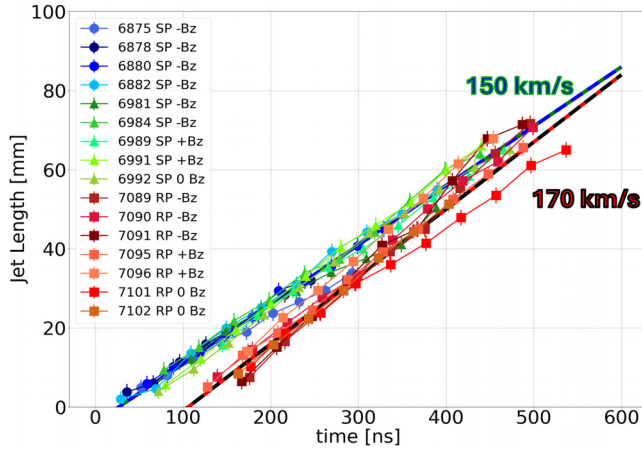


FIG. 7. Plot of jet length as a function of time for selected jets in the first three experimental shot campaigns.

to each electrode. If the valve next to the load hardware (see Fig. 2) provides more gas compared to the valve below the load hardware, the expected result would be a longer pinch time in RP configuration (higher density on axis near $z = 0$) and a more rapid expansion of the jet (reduced density from annular slit) and vice versa for SP configuration. This is consistent with our observations. Extended MHD effects may also contribute to these differences; however, to rigorously test this hypothesis, future studies must carefully control for confounding factors such as variations in the density distribution. Addressing these issues will be a focus of future research. For now, we note that the RP configuration produces higher quality jets (in terms of collimation, stability, and length) more consistently than the SP configuration.

To measure the current within the jets, $\vec{B}$ probes were positioned outside of the central column. These probes are based on the design described in Ref. 64 but were encased in an additional alumina ceramic

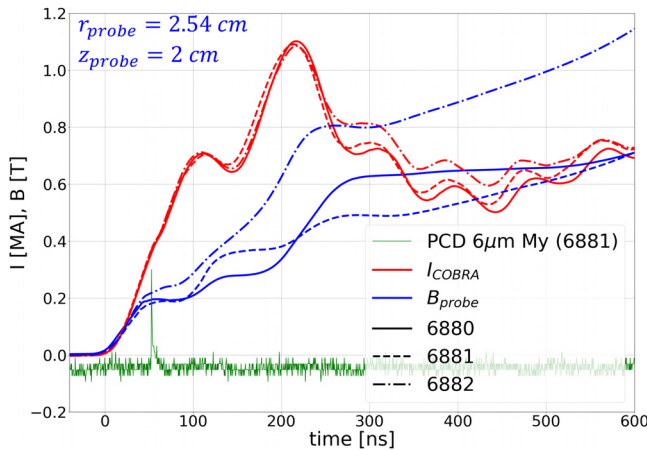


FIG. 8. Integrated $\vec{B}$ probe measurements compared to the COBRA current for three shots in the first experimental campaign. The probe was oriented to measure the azimuthal component of the magnetic field (B_θ) at a radius of 2.54 cm and a height of 2 cm. A representative PCD signal is shown for shot 6881 indicating the time at which the base of the jet first pinches.

tube to improve survivability over multiple shots. Figure 8 displays the integrated $\vec{B}$ signals for three shots in the first experimental campaign, taken at a radius of 2.54 cm and a height of 2 cm, oriented to measure the azimuthal magnetic field B_θ . The probe's position is shown below the two radial tungsten pins in Figs. 2(b) and 6(a). Initially, the azimuthal magnetic field increases proportionally to the driver current until the base of the jet pinches and a short burst of x-rays are detected. At this point ($t \approx 60$ ns), the measured azimuthal field decouples from the driver current. For shots 6880 and 6881, the field then strengthens incrementally in a stepwise fashion, with increases around $t \approx 100$ ns and $t \approx 200$ ns before stabilizing around 250 ns. For shot 6882, a steadier increase in B_θ is observed. Late time growth in the integrated $\vec{B}$ signal is also observed for $t \geq 300$ ns; however, it is believed that the signal at this stage is no longer indicative of the current flowing within the jet. The peak azimuthal magnetic field at the probe radius, therefore, ranges from 0.5–0.8 T, which corresponds to an axial jet current of 64–100 kA, or $\leq 10\%$ of the peak driver current. Explanations for this discrepancy include the possibility that a significant fraction of the return current flows within the radius of the $\vec{B}$ probe (e.g., a dominant $m = 1$ field configuration with intertwined forward and reverse currents^{24,26}) or that much of the current is flowing through the safety wire. The stepwise increase in B_θ is also interesting and may be attributed to a helical kink instability, which alters the distance between the probe and the current-carrying central column, or alternatively, to some other relaxation event that allows more current to couple into the jet. To resolve these questions, angularly resolved $\vec{B}$ measurements taken in subsequent shot campaigns are currently being analyzed. Faraday rotation imaging will also be used to provide detailed 2D mappings of the azimuthal field distribution. The results of these investigations will be presented in future work.

The internal jet current can also be estimated from Figs. 6 and 7 using a theoretical model, which describes the dependence of the average axial velocity on jet current.⁶⁵ In this model, the flow is driven by an axial gradient in toroidal magnetic field resulting from a flared jet with a radius described by $a(z) = a_0 e^{\kappa z}$. With modest simplifying assumptions, the average axial velocity predicted by this scenario is⁶⁵

$$u_z(r=0, z=L) \sim \frac{1}{\pi a_0} \sqrt{\frac{\mu_0 \kappa L}{\rho_0}} I_0, \quad (2)$$

where μ_0 is the vacuum permeability, and ρ_0 and I_0 are the mass density and average current at the base of the jet. For our jet near peak machine current ($t \approx 220$ ns), $L \approx 30$ mm, $a \approx 1.5$ mm, $\rho_0 \approx 0.06$ kg m⁻³, $u_z \approx 150$ –170 km/s, and $\kappa \sim 100$ m⁻¹. This corresponds to an average internal jet current of $I_0 \approx 89$ –101 kA, which is consistent with the $\vec{B}$ probe measurements presented above. Subsequent ideal MHD simulations⁶³ have corroborated this acceleration model while reproducing many of the features that we observe, such as a uniform expansion velocity of the leading edge and an axial velocity gradient that we identify below. The difference in expansion velocity between SP and RP configurations may also be explained by the $\rho_0^{-1/2}$ scaling in Eq. (2). The RP configuration is expected to have a higher ρ_0 near $z = 0$, consistent with its longer pinch time. However, given the nature of the acceleration mechanism and defining the base of the jet as the point where the flaring begins, Eq. (2) predicts that beyond $z \sim 10$ mm, the on-axis density for the RP configuration is $\sim 80\%$ of that in the SP configuration. This prediction can be directly tested in

future PLIF measurements of the neutral gas density profile for both configurations.

Line integrated electron density was measured in each shot campaign using Mach–Zehnder interferometry with a 140 mJ, 150 ps, ND:YAG laser at 532 nm. Figure 9 shows the analyzed interferometry data for various jets in each campaign for different initial conditions. The results reveal that the jets are quite inhomogeneous with complex internal structure consisting of clumps, knots, and filaments. Jet radii range from about 1–2 mm, which, for a mean line density of $2 \times 10^{18} \text{ cm}^{-2}$, gives local electron densities of the order of 10^{19} cm^{-3} . There is substantial variation in jet structure between the shots. For shot 6875 (SP, $-B_z$), close examination reveals two distinct filaments at $z \approx 7.5 \text{ mm}$, which intersect each other multiple times along the length of the jet. XUV images from an orthogonal direction also reveal the intersecting filaments, confirming that they are intertwined like a double-helix. Similar filaments can be seen in the visible self-emission images for shot 6880 in Fig. 6(a) at 225 ns, which resembles the classic “Eiffel tower” or “λ” morphology observed in many solar coronal jets.^{66,67} For shot 6991 (SP, $+B_z$), the line integrated electron density reveals a clumpy jet and any filaments, if present, are not well resolved. Shot 7095 (RP, $+B_z$), on the other hand, produced a more uniform, wider jet, with multiple pronounced helical filaments. The pitch angle of the filaments is quite steep, such that if they are aligned with the magnetic field, the poloidal field must dominate over toroidal field produced by axial current within the jet. This seems unlikely given the initial strength of the poloidal field, and we note that anomalously steep helical perturbations have also been observed in solid liner implosions⁶⁸ and gas-puff z-pinch implosions,^{69–72} where various mechanisms such as axial magnetic flux compression by force-free currents in the low density corona⁷³ or a Hall interchange instability⁷⁴ have been proposed to account for the observed structures. Shots without an initial poloidal field (no magnet) also produced highly collimated, stable jets with fascinating internal structure. For shot 7100 (RP, $0 B_z$) for example, the jet exhibits complex small-scale features between $z = 11\text{--}20 \text{ mm}$, after which two large scale helical filaments emerge and extend outside of the image frame. It is not yet clear whether the helical structure is formed at the source and then

ejected from the electrodes or whether it emerges via some relaxation process operating in the flow (i.e., an inverse cascade). Future time resolved measurements and computer simulations will seek to answer this question.

To measure jet temperature and velocity profiles, optical Thomson scattering was performed in the collective scattering regime with a focus on the ion acoustic wave spectral feature. Scattering spectra were modeled and fit using a collisionless theoretical spectral density function.⁷⁵ The analysis method and fit parameters are the same as in Ref. 76, with the exception that ion and electron temperatures are assumed to be equal and the electron density for each collection fiber is estimated based on simultaneous interferometry measurements and the relative intensity of the scattering signal. Deviation from thermal equilibration is accounted for in the Monte Carlo error analysis, which additionally allowed for a 50% standard deviation of the ion to electron temperature ratio. For the first experimental shot campaign, the Thomson laser bisected the chamber center at a height of 14.5 mm from the electrode plane. For the second and third campaigns, the laser was at a height of 20 mm. Figure 10 shows the Thomson scattering geometry used in the experiments. For each campaign, scattering spectra were collected from two scattering angles in the XY plane at $\pm 90^\circ$ using a bifurcated fiber bundle, allowing for determination of the two orthogonal velocity components in that plane. For several shots, spectra were also collected from the $\hat{z}$ direction, enabling the calculation of all three orthogonal velocity components.

The 1D flow profiles obtained from Thomson scattering are often difficult to interpret, with significant shot-to-shot variation. In Fig. 11, however, we present some of the more “well-behaved” flow profiles that reveal trends seen throughout the dataset. These trends include indications of azimuthal rotation (opposite sign of v_{θ} on either side of the jet axis), radial inflow ($+v_{kr}$ at $-r$ position and vice versa), and axial shear flows with a hollow profile. Velocity magnitudes for components in the XY plane range from $\sim 0\text{--}40 \text{ km/s}$ while axial velocities range from $\sim 20\text{--}80 \text{ km/s}$. Temperatures generally peak on axis at around $\sim 30\text{--}50 \text{ eV}$; however, temperatures approaching $\sim 120 \text{ eV}$ have been measured for shots without an initial poloidal field. In some cases, multiple peaks in the temperature profile are observed, possibly indicating intertwined filaments. Laser heating of the plasma via inverse bremsstrahlung is a concern, particularly for the more

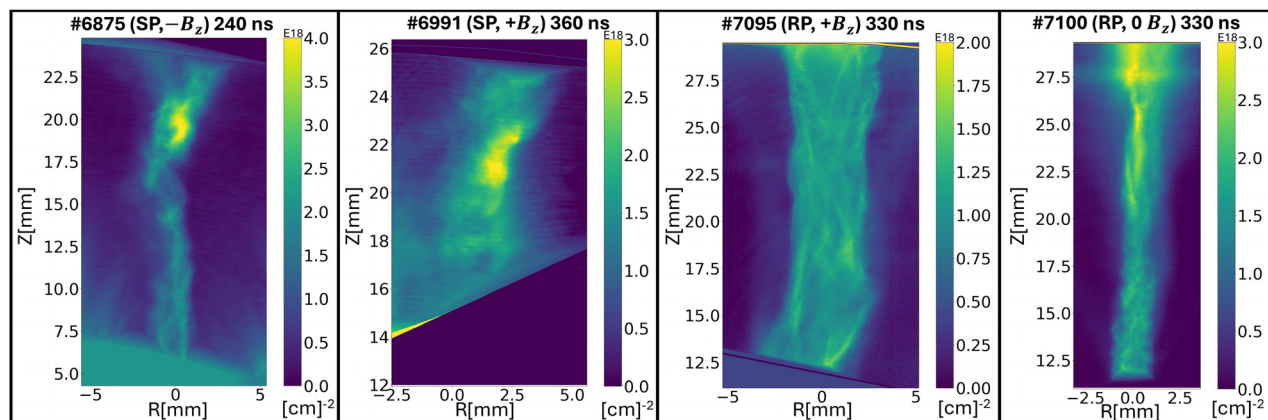


FIG. 9. Line integrated electron density for various plasma jets in each shot campaign.

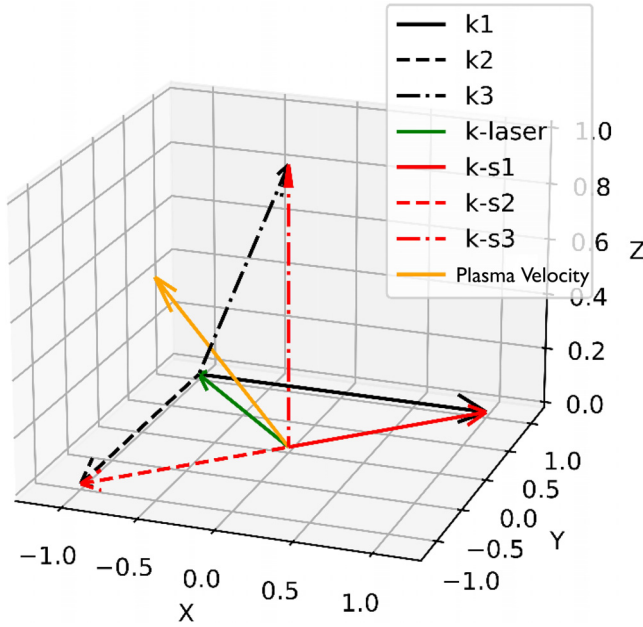


FIG. 10. Thomson scattering geometry showing the laser wave vector (k -laser), three scattering wave vectors (k -s1, k -s2, k -s3), and the resulting measurement wave vector (k_1 , k_2 , k_3) equal to the difference between the scattering and laser wave vectors. Simultaneous measurements along k_1 and k_2 enable reconstruction of a velocity vector projected in the XY plane while all three components enable reconstruction of the full 3D vector.

dense jets. In subsequent campaigns, the laser energy will be reduced to ensure any contribution from the laser is mitigated.

For temperatures of 30–50 eV and electron densities of 10^{19} cm^{-3} , the mean ionization state is expected to be ~ 8 .⁷⁷ Assuming an

effective adiabatic index of ~ 1.4 ,⁷⁸ the resulting internal ion sound speed is approximately 30–40 km/s, which is equivalent to the Alfvén velocity (v_a) for a $\sim 10 \text{ T}$ internal magnetic field (consistent with B measurements). This gives a ratio of thermal to magnetic pressure, $\beta \approx 1$, as is generally expected near the launch region of YSO jets. If temperatures measured near the edge of the jet ($\lesssim 10 \text{ eV}$) are representative of the ambient plasma temperature, the external sound speed is approximately 10 km/s. The sustained axial flow at the Thomson scattering measurement height is, therefore, internally transonic or weakly supersonic but hypersonic with respect to the surroundings. Meanwhile, the velocity of the tip of the jet is approximately hypersonic with respect to the internal flow (assuming similar temperatures) and strongly hypersonic with respect to the external medium. This apparent increase in velocity from the base of the jet to the tip has also been observed in the proper motion of YSO jets⁶² as well as MHD simulations⁶³ and suggests an internal acceleration mechanism such as the one described in Ref. 49. Examination of the axial flow velocity profile for shot 6991 reveals a radial shear, $\partial v_z / \partial r \approx 10^7 \text{ s}^{-1}$, which exceeds the kink instability stabilization threshold of $0.1 k v_a \approx 2 \times 10^{6,79,80}$ by nearly an order of magnitude for the worst case instability with a wave number $k = \pi/d$, where d is the jet diameter. These strong helical shear flows are predicted to contribute to the stability of the plasma column,⁴⁶ considering the jet is expected to be classically unstable with $q(a) = 2\pi a B_z(a) / L B_0(a) < 1$, where q is the safety factor.⁴⁸ Interestingly, the axial shear profile is hollow, with the highest velocity near the jet edges. This is inconsistent with the model presented in Ref. 49, which predicts that the axial velocity (driven by an axial pressure gradient) should be peaked on axis. However, at the time and location of our measurement, the bulk jet is well collimated, so any axial pressure gradient should be small. The observed flow profile is, therefore, more consistent with plasma being driven onto the jet from the periphery.

Based on the experimental measurements in the first three shot campaigns, Table I compares the COBRA jet plasma parameters to

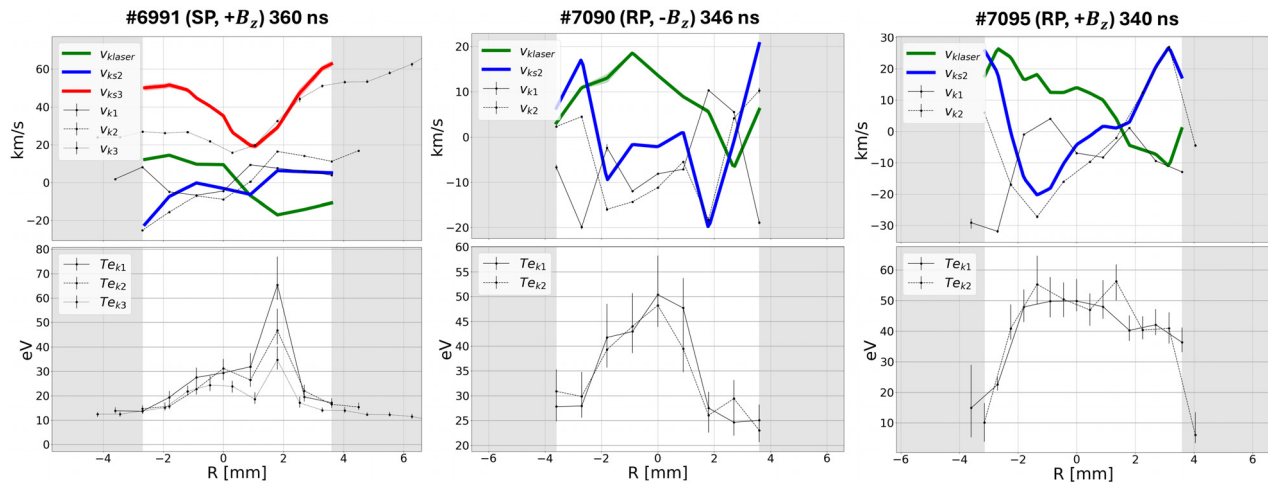


FIG. 11. Thomson scattering measurements for three jets in campaigns 2 and 3. The top panels show plasma velocity along the measurement wave vectors for each linear fiber bundle and the resulting calculated flow velocities along the laser wave vector, k_{s2} , and k_{s3} (for shot #6991) where the measurements overlap. Assuming a cylindrical jet centered on axis, these three orthogonal directions correspond to the radial, azimuthal, and axial direction, respectively. The lower panels show the corresponding jet temperature measured with each fiber bundle. Note that while the laser intercepts the chamber center, the jet column may be offset from the axis at the measurement height.

empirically grounded order of magnitude approximations for YSO jets as well as jets produced on the MOCHILabjet experiment.^{45,46} Despite large differences in dimensional parameters, COBRA jets satisfy the MHD scaling relations for both astrophysical and planar plasma gun jets to within an order of magnitude, while the dimensionless parameters such as Reynolds, magnetic Reynolds, and Péclet numbers are sufficiently large to justify the applicability of such scaling laws.⁵⁰ A Reynolds number of $\sim 10^5$ suggests the jets are turbulent, a hypothesis that appears to be supported by interferometry data. Magnetic Reynolds and Péclet numbers are also much larger than unity (advection dominated) with comparable magnitudes, as may be expected near the launching region as thermal and magnetic pressures remain balanced ($\beta \approx 1$).

IV. SUMMARY AND FUTURE WORK

Predicated on the assumption that the underlying physical processes responsible for the formation, collimation, and stability of astrophysical jets is scale invariant and robust (i.e., occurring over a broad range of physical parameters), a new experimental platform for the 1-MA COBRA generator has been developed to produce and study these outflows. This novel platform combines the approach of previous medium energy density planar plasma gun experiments^{45–49} with the pulsed-power and diagnostics capabilities of the COBRA facility. Using symmetric gas-puff injection instead of traditional solid targets such as wire arrays or foils, there is no initial ablation stage, allowing magnetized jets to form earlier in the current pulse and be driven longer without disruption. Poloidal magnetic fields, required in many theoretical models of astrophysical jet formation, can be easily provided through the addition of a powerful permanent magnet to the central electrode. The new platform can also couple with an existing polarity reversing convolute to investigate extended MHD effects.

Preliminary observations following the first three experimental campaigns reveal that the jets formed by this new platform are remarkable and unique for their length, degree of collimation, and stability. Evolving over more than 500 ns, the jets achieve lengths in excess of 7 cm and radii on the order of 1.5 mm with little to no divergence. The expansion velocity of the jets is hypersonic, while sustained flows near the base of the jet are transonic or weakly supersonic, with indications of rotation and axial shear. The magnitude of the axial shear flow is more than sufficient to stabilize current driven kink instabilities for the fastest growing axial modes.⁷⁹ The jets are quite dense compared to their surroundings, with electron densities on the order of 10^{19} cm^{-3} and complex internal structure consisting of clumps, knots, and filaments. Jet temperatures are generally peaked on axis at around 30–50 eV; however, temperatures approaching $\sim 120 \text{ eV}$ have been measured for shots without an initial poloidal field. The measured parameters demonstrate that COBRA jets align well with MHD scaling relations for YSO jets and jets from earlier planar plasma gun experiments. These systems share similar boundary conditions, stability characteristics, and morphology, suggesting they evolve similarly when described using appropriate dimensionless variables. The consistent behavior across such vastly different parameter spaces further validates the scaling laws and suggests that these jets are effective laboratory analogs for YSO jets.

Currently, work is ongoing to support these experiments through extended MHD modeling using the PERSEUS code.⁸⁵ After benchmarking the results with experimental data, the simulations will be used to investigate in detail the injection and transport of relative canonical

helicity³⁴ and the relaxation dynamics governing flows and magnetic fields in these driven equilibria. We will also seek to better characterize the neutral gas-puff density distribution of the load prior to a shot in order to isolate and identify XMHD effects from other confounding influences. Laser energy for Thomson scattering will also be reduced in subsequent shot runs to eliminate the possible influence of laser heating through inverse bremsstrahlung. We will also seek to investigate the effect of different gas species, different puff density distributions, as well as the role of the safety wire on jet dynamics and stability.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

E. S. Lavine: Conceptualization (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (equal); Methodology (equal); Project administration (lead); Supervision (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **D. A. Lund:** Formal analysis (supporting); Investigation (equal); Methodology (equal); Visualization (supporting); Writing – review & editing (supporting). **E. Freeman:** Investigation (supporting); Methodology (supporting); Writing – review & editing (supporting). **W. M. Potter:** Investigation (supporting); Methodology (supporting); Writing – review & editing (supporting). **C. E. Seyler:** Supervision (supporting); Writing – review & editing (supporting). **B. Kusse:** Funding acquisition (supporting); Project administration (supporting); Supervision (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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